# Generation drawing ports

Current continuation: after approved prerequisites through source36, source38 is prepared in the separate mark geometry lane. The original approved bookmark and evidence below remain preserved historical source15/16 records; later absence/deferment statements describe their old base.

## Scope and source dispositions

This record describes the isolated drawing lane based on approved prefix `74cebf6c`, workspace and bookmark `generation-drawing-ports`, change `snvmvumx`. Astra coordinator round 3 approved its cumulative tip `98cbf5b8`, following the independent runtime/layout/packaging reviews. The integration imports the cumulative patch at `8430e701`. Its combined staged-transplant review remains pending. The historical source38 prerequisite discussion below describes the isolated lane base: the current source17/18 integration adds the shared mark engine and retains Tiptoe attribution. Source38 remains unimplemented. The accepted plan and its WP2 manifest remain authoritative. The structured receipt records exact source identities, changed files, validation and remaining gates.

The already-approved notices change `00df0507` is replayed with its original license texts and common root-license selector. Its original workspace and bookmark are unchanged. Clarabel remains the solver; NetworkX-derived native seed code remains. Both notices accompany the embedded and packaged assets. The development sync recipe now also copies Kurvst's package license.

Source15 removes the unused GPL-linked `ec-layout.wasm`, its combined license, Emscripten build and packaging references. The native reference engine is preserved. The generated drawing recipe no longer names the absent plugin, and the model style no longer imports the obsolete `mi` helper. Supplied-geometry render coverage is retained and additionally asserts that the removed artifacts are absent. The native/static notebook display and optional widget contract from merged #209 are untouched.

Source16 uses the exact manifest identity `9e737a18274991e8dc0542f777a6063acaf72fe8`. Placement search ranks painted-text contacts before clearance costs, excludes contained arrow-footprint pieces, and retains the best state across annealing and descending passes. A second walk that cannot increase contacts is considered when the first still has contacts. Geometry is intentionally a binary64 drawing boundary, independent of precision-generic physics. The new contact policy is a named internal enum; new score types include debug support and imported names follow Rust policy.

Source38 is deferred, not partially claimed. This base has no `crates/kurvst/src/marks.rs`, `crates/kurvst/typst/src/mark.typ`, shared mark exports, or Tiptoe license copies. Its native SVG renderer still owns its smaller local `svg/marks.rs`. The shared engine and its consumers arrive through the much larger source17 transplant, outside this slice's authorization. Replaying source38 alone would either fail or introduce an unused parallel engine. Resume only after the independently reviewed source17/18 transplant and its remaining prerequisites through port36. Then replace geometry and remove Tiptoe notices only once their owning code is replaced. No historical WASM, SVG or PDF output is copied. There is consequently no changed mark geometry or mark manual in this slice to certify.

## Round 1 metadata correction

Astra round 1 requested changes on `d5862a8e`: the delivered Typst package still declared the removed plugin's GPL clause. The correction declares exactly `MIT AND Apache-2.0 AND BSD-3-Clause`, retaining all Clarabel and NetworkX notice texts. Regression assertions check the parsed metadata in the actual source archive and API-extracted package, while the Nix bundle check asserts

its delivered manifest expression. The affected bundle is rebuilt with the canonical recipe. Prior verified code/render evidence remains preserved; the correction and its artifact identities are recorded in the fix receipt. This fix subsequently passed independent review; WP11 is not complete.

## Round 2 registry correction

Astra round 2 closed the original license P2. Its new P2 found that the drawing style migration registry still hashed the pre-notice-test `templates.rs` file. The registry owner hashes the whole file because the scope has no end anchor. Comparison against approved prefix `74cebf6c` confirms that all production source before the test module is byte-identical; only package-layout license assertions changed. A dated test-only no-architecture-impact attestation now accompanies the digest calculated from the actual current file. The round 2 fix receipt records validation of every registry scope whose source changed in this drawing slice, using the owner's exact anchor slicing and Git blob hashing semantics. No runtime source, package metadata, notice or rendered asset changes in this fix. Astra coordinator round 3 closed this registry finding at `98cbf5b8`; WP11 and the final artifact gates remain incomplete.

## Validation and remaining gates

The following geometry and render results are preserved pre-fix evidence, also independently confirmed in Astra round 1. The metadata fix repeats only the affected packaging, bundle, formatting and note checks; its receipt distinguishes those results from the earlier full drawing validation.

All 245 Kurvst/Linnest geometry and layout tests pass; the fixture-rewriting test remains skipped. Four packaging regressions, two final-bundle Clinnet notice regressions and three actual Typst seed/layout regressions pass. All 38 Clinnet tests, including public Typst rendering, pass. Scoped Clippy with denied warnings and workspace formatting pass. The final Nix WASM check passes, and an independent Nix rebuild matches its outputs. All 34 website SVGs reproduce byte-for-byte; only the two expected `epem-ttbar-cut` assets change. Both themes were rendered to PNG and visually inspected for label and arrow placement. The source16 manual PDF is rebuilt with the HTML feature required by its existing `html.frame` calls and matches a second timestamp-fixed render. The receipt records the broader Clinnet render suite and exact commands, hashes and limits. Existing useful galleries are preserved. All changed drawing outputs are rebuilt from this checkout.

Standalone Linnet packaging is absent on this base. Its eventual restoration must retain `license-files = ["LICENSE", "licenses/*"]`, include Clarabel and NetworkX notices and verify the final artifact metadata and bytes. Fixture wheels certify notice delivery only; they do not certify full native or Pyodide runtime wheels. Full native/Pyodide host and external Community artifact checks remain final-integration gates. Solver replacement, kkumutpy, particle(momentum), gallery/API ports, publication and historical cleanup remain outside this slice.